CS228 Logic for Computer Science
Exercise Problem Set 7

Part 1

1. Prove or disprove each of the following equivalences. Here $Q \in \{\forall, \exists\}$, and unless stated otherwise, $x \notin free(\psi)$.

   If an equivalence is true, prove it semantically, assuming the universe is non-empty (part (h) looks at what changes when it may be empty). You may use Exercise 13.2 from Lecture 13: if environments α, α' agree on $free(\varphi)$, then $M, \alpha \models \varphi$ iff $M, \alpha' \models \varphi$. (We call this the *Coincidence Lemma*.) You may also give an ND proof if you wish.
   If it is false, give a structure with a non-empty universe (and an environment, if there are free variables) under which the two sides differ.

   (a) $(Qx\,\varphi) \wedge \psi \;\equiv\; Qx\,(\varphi \wedge \psi)$

   (b) $(Qx\,\varphi) \vee \psi \;\equiv\; Qx\,(\varphi \vee \psi)$

   (c) $\psi \to Qx\,\varphi \;\equiv\; Qx\,(\psi \to \varphi)$

   (d) $(Qx\,\varphi) \to \psi \;\equiv\; Qx\,(\varphi \to \psi)$

   (e) $Qx\,\varphi \;\equiv\; Qy\,\varphi[y/x]$, where y does not occur in φ

   (f) $(\forall x\,\varphi) \wedge \psi \;\equiv\; \forall x\,(\varphi \wedge \psi)$, where now $x \in free(\psi)$

   (g) $\forall x\,\exists y\,R(x,y) \;\equiv\; \exists y\,\forall x\,R(x,y)$

   (h) Suppose we now allow the empty universe $U^M = \varnothing$. Then there is no environment, so consider only sentences: every sentence $\forall x\,\chi$ is true and every sentence $\exists x\,\chi$ is false. Which of (a)–(g) change status?

2. Recall that a formula is *rectified* if no two quantifiers use the same variable, and no variable occurs both bound and free. Rectify each of the following formulas, i.e., rename bound variables to obtain an equivalent rectified formula.

   (a) $\forall x\,P(x) \wedge \exists x\,Q(x)$

   (b) $\forall x\,(P(x) \to \exists x\,Q(x,y))$

   (c) $\exists x\,P(x,y) \vee \forall y\,\exists x\,R(x,y)$

   (d) $P(x) \wedge \forall x\,\big(Q(x) \to \exists x\,R(x,x)\big)$

   (e) $\neg\big(\neg P(x) \vee \forall x\,Q(x)\big) \wedge \big(\exists x\,P(x) \to \forall x\,R(x)\big)$

   (f) For each formula above, list the free and bound variables, before and after rectification. Why can a free variable never be renamed?

3. Which of the following formulas are in prenex normal form? Convert those that are not into rectified prenex form (RPF), by rectifying them first where needed (Problem 2).

   (a) $\forall x\,(P(x) \to \exists y\,Q(y,x))$

(b) $\forall x\, P(x) \to \exists x\, Q(x)$

(c) $\neg\forall x\, \exists y\, R(x,y) \;\vee\; \forall z\, S(z)$

(d) $\neg\big(\neg P(x) \vee \forall y\, Q(y)\big) \wedge \big(\exists z\, P(z) \to \forall w\, R(w)\big)$ (the rectified formula from Problem 2(e))

(e) Convert (b) in two different ways so that the resulting quantifier prefixes differ. Are the results equivalent? Why or why not?

(f) Give a formula with no free variables for which pulling the quantifiers to the front *without* first rectifying changes its meaning. (Problem 1(f) already shows what goes wrong with a free variable.)

4. Skolemize each of the following sentences, i.e., convert it to Skolem Normal Form.

(a)–(c) The prenex forms of the sentences in Problem 3(a)–(c).

(d) $\forall x\, \exists y\, \forall z\, \exists w\, R(x,y,z,w)$

5. Consider the vocabulary $\mathcal{V} = \{0, +, \leq, =\}$ and the $\mathcal{V}$-structure $\mathcal{N} = (\mathbb{N}, 0, +, \leq, =)$ with universe $\mathbb{N} = \{0, 1, 2, \dots\}$ and the usual interpretations. Answer the following.

(a) Check whether each sentence is true in $\mathcal{N}$:

i. $\forall x\, \exists y\, (x \leq y)$

ii. $\exists x\, \forall y\, (y \leq x)$

iii. $\forall x\, \forall y\, \exists z\, (x + z = y \vee y + z = x)$

(b) Which of the sentences in (a) change truth value in $(\mathbb{Z}, +, \leq, 0)$? In $(\mathbb{Q}, +, \leq, 0)$? In $(\{0,1,2\}, \max, \leq, 0)$?

(c) Write a formula with one free variable defining "x is even".

(d) Is $\forall x\, \exists y\, (x \leq y \wedge \neg(x = y))$ true in $\mathcal{N}$? Skolemize it and give an interpretation of the Skolem function f in $\mathcal{N}$ under which the matrix is true.

(e) Do the same for $\exists x\, \forall y\, (x \leq y)$: Skolemize it, and give an interpretation of the Skolem constant c in $\mathbb{N}$ under which the Skolemized formula is true (with the other symbols interpreted as in $\mathcal{N}$).

Part 2

6. A normal form is *canonical* if for every formula φ there is a *unique* formula in that normal form which is equivalent to φ.

(a) Is the Rectified Prenex Form (RPF) canonical? Why or why not?

(b) Is the Skolem Normal Form (SNF) canonical? Why or why not?

Hint: Revisit Problem 3(c) and pull out the quantifiers in a different order.

7. In this problem we show that conversion to RPF preserves equivalence, while Skolemization only preserves satisfiability. For a term t, write $\overline{\alpha}(t)$ for the value of t in M under the environment α (as in Exercise 13.2).

(a) *(Substitution Lemma)* Let t be a term that is free for z in ψ. Prove that for every structure M and environment α,

$$M, \alpha \models \psi[t/z] \quad \text{iff} \quad M, \alpha[z \mapsto \overline{\alpha}(t)] \models \psi.$$

Hint: first show $\overline{\alpha}(s[t/z]) = \overline{\alpha[z \mapsto \overline{\alpha}(t)]}(s)$ for every term s, then use structural induction on ψ, as in the proof of the Coincidence Lemma. Where do you need that t is free for z?

(b) Show that a formula and its RPF are equivalent.

(c) Let $\varphi := \forall y_1 \cdots \forall y_n \, \exists z \, \psi$ be a rectified sentence, and let $\varphi' := \forall y_1 \cdots \forall y_n \, \psi[f(y_1, \ldots, y_n)/z]$, where f is an n-ary function symbol not occurring in φ (a constant if $n = 0$). Using (a), show that φ and φ' are *equisatisfiable*: φ is satisfiable iff φ' is. Conclude that a sentence and its SNF are equisatisfiable.

(d) Show that a sentence and its SNF need not be equivalent. *Hint:* Problem 4(b).

(e) What does this imply about the uniqueness of Skolem functions?

8. A *universal sentence* has the form $\forall x_1 \cdots \forall x_n \, \psi$ with ψ quantifier-free. Given $\mathcal{V}$-structures $\mathcal{M}$ and $\mathcal{N}$, we say $\mathcal{N}$ is a *substructure* of $\mathcal{M}$ if $U^{\mathcal{N}} \subseteq U^{\mathcal{M}}$ is non-empty, contains $c^{\mathcal{M}}$ for every constant c, and is closed under $f^{\mathcal{M}}$ for every function symbol f; and $\mathcal{N}$ interprets every symbol as $\mathcal{M}$ does, restricted to $U^{\mathcal{N}}$ (so $c^{\mathcal{N}} = c^{\mathcal{M}}$, $f^{\mathcal{N}}$ is $f^{\mathcal{M}}$ restricted to $U^{\mathcal{N}}$, and $P^{\mathcal{N}} = P^{\mathcal{M}} \cap (U^{\mathcal{N}})^k$ for each k-ary predicate P).

(a) Show that universal sentences are preserved under substructures: if $\mathcal{M} \models \theta$ for a universal sentence θ and $\mathcal{N}$ is a substructure of $\mathcal{M}$, then $\mathcal{N} \models \theta$.

(b) Let $\mathcal{V}$ contain no function or constant symbols (only predicate symbols). Show that every satisfiable universal $\mathcal{V}$-sentence has a model whose universe has exactly one element.

(c) Let $\mathcal{V}$ be as in (b), and contain a binary predicate symbol. You may assume *Church's theorem*: there is no algorithm that decides, given a $\mathcal{V}$-sentence, whether it is satisfiable. Prove that there is no algorithm T that converts every $\mathcal{V}$-sentence φ into a universal $\mathcal{V}$-sentence $T(\varphi)$ such that φ is satisfiable iff $T(\varphi)$ is satisfiable. What does this show about the fresh symbols introduced by Skolemization?

(d) Consider $\varphi := \forall x \, \forall y \, \exists z \, (\neg(z = x) \wedge \neg(z = y))$. Show that φ is satisfiable. Then show that for every term t whose only variable is x (or which has no variables), the sentence $\forall x \, \forall y \, (\neg(t = x) \wedge \neg(t = y))$ is unsatisfiable, and likewise for terms whose only variable is y. Conclude that a Skolem term for z must in general depend on all the universally quantified variables before it.

9. For a sentence φ, write φ^S for its SNF (i.e., Skolemize an RPF of φ), and define its *Herbrand form* $\varphi^H := \neg\big((\neg\varphi)^S\big)$. Dually to Skolemization, this eliminates the *universal* quantifiers of φ using fresh function symbols, leaving only existential ones.

(a) Compute φ^H for the formula $\varphi = \exists x \, \forall y \, (P(x) \to P(y))$ and verify directly that φ^H is valid.

(b) Prove that φ is valid iff φ^H is valid.

(c) Show that $\varphi \models \varphi^H$ in general, but that $\varphi^H \not\models \varphi$ already for $\varphi = \forall x \, \exists y \, R(x, y)$.

(d) Show that the Herbrand form does *not* preserve satisfiability: find a satisfiable φ^H whose φ is unsatisfiable.

10. Recall the Compactness Theorem from Lecture 16: if every finite subset of a (possibly infinite) set Γ of sentences is satisfiable, then so is Γ. For $n \geq 1$, let

$$\varphi_n := \exists x_1 \cdots \exists x_n \bigwedge_{1 \leq i < j \leq n} \neg(x_i = x_j).$$

(a) Show that $M \models \varphi_n$ iff U^M has at least n elements.

(b) Let ψ be a sentence such that for every $n \geq 1$ there is a structure M_n whose universe has at least n elements, with $M_n \models \psi$. Show that ψ has a model whose universe is infinite. (This is the Löwenheim–Skolem theorem.)

(c) Show that there is no set Σ of sentences whose models are exactly the structures with a *finite* universe. (So "the universe is finite" is not FO-definable.)